Charotar University of Science and Technology
Chandubhai S Patel Institute of Technology

Third Semester of B. Tech (CE/IT/CSE/AIML)

CEUE203 Object Oriented Programming

Practical Examination – Problem Definitions (Units 1, 2 and 3)

SET B

Problem 1: Stationery Shop Receipts (Units 1 & 2)

The college stationery shop issues a receipt number to every valid purchase and keeps a running total of revenue. Write a program that reads purchases, prints a receipt line for each valid purchase and prints a summary at the end.

PRICE LIST (FIXED)

Code	Item	Price (₹)
N	Notebook	45
P	Pen	10
M	Marker	25
F	File	30

TASK

- A class **Product** with **private final** fields `name` (String) and `price` (int), a constructor `Product(String name, int price)` that uses **this**, and getters `getName()` and `getPrice()`.
- A method **static Product findProduct(char code)** that uses a **switch expression** (arrow form) to return the matching Product object, or `null` for any other code.
- A class **Purchase** with:
    - **private** fields `receiptNo` (int), `product` (Product) and `quantity` (int);
    - **static** variables `nextReceipt` (starting at 201), `totalPurchases` and `totalRevenue`;
    - a constructor `Purchase(Product product, int quantity)` that uses **this**, assigns the next receipt number and updates both static totals;
    - `getAmount()`, which returns price × quantity;
    - **static** methods `getTotalPurchases()` and `getTotalRevenue()`;
    - an overridden **toString()** in the format shown in the Output Format.
- A purchase is **invalid** if the code is not on the price list **or** the quantity is ≤ 0. Do not create a Purchase object for an invalid purchase, and do not use up a receipt number for it.

INPUT FORMAT

Line 1: an integer N, the number of purchases.
Next N lines: a code (a single uppercase letter) and a quantity (an integer), separated by a space.

OUTPUT FORMAT

For each valid purchase: `Receipt <receiptNo>: <ProductName> x <quantity> = <amount>`
For each invalid purchase: `Invalid item: <code> <quantity>`
Last line: `Total receipts: <count>, Total revenue: <amount>`

CONSTRAINTS

$1 \le N \le 20$
$-100 \le$ quantity ≤ 100

SAMPLE INPUT 0

```
5
N 2
P 5
Z 1
M 0
F 3
```

SAMPLE OUTPUT 0

```
Receipt 201: Notebook x 2 = 90
Receipt 202: Pen x 5 = 50
Invalid item: Z 1
Invalid item: M 0
Receipt 203: File x 3 = 90
Total receipts: 3, Total revenue: 230
```

Explanation: Z is not on the price list and M has quantity 0, so neither gets a receipt number. The receipt numbers therefore stay continuous: 201, 202, 203.

SAMPLE INPUT 1

```
4
M 4
N 1
F 2
P -3
```

SAMPLE OUTPUT 1

```
Receipt 201: Marker x 4 = 100
Receipt 202: Notebook x 1 = 45
Receipt 203: File x 2 = 60
```

```
Invalid item: P -3
Total receipts: 3, Total revenue: 205
```

Problem 2: Notice Cleaner (Unit 3)

Students type notices quickly, so they often contain extra spaces and mixed letter case. Write a program that cleans one line of notice text and reports some statistics about it.

TASK

Remove leading and trailing spaces and split the line into words. Words may be separated by **one or more spaces**.

- **Cleaned sentence:** write the whole sentence in lowercase, except that the first letter of the first word is uppercase, and join the words with single spaces. Build it using a **StringBuilder**.
- **Word count:** the number of words.
- **Shortest word:** the shortest word in lowercase. If two or more words have the same shortest length, print the **first** one.
- **Keyword count:** how many words are equal to the given keyword, **ignoring case**. Only whole words count, so *theory* does not match *the*.
- **Vowel count:** the total number of vowels (a, e, i, o, u in either case) in the original line, counted using a loop over its characters.

INPUT FORMAT

Line 1: the notice sentence. It may contain leading, trailing or repeated spaces, and contains only letters and spaces.
Line 2: the keyword (a single word).

OUTPUT FORMAT (EXACTLY FIVE LINES)

```
Sentence: <cleaned sentence>
Words: <count>
Shortest word: <word>
Keyword count: <count>
Vowels: <count>
```

CONSTRAINTS

The sentence contains at least 1 word and at most 100 characters.
The keyword contains only letters.

SAMPLE INPUT 0

```
  spring SEMESTER exam is   near and exam is TOUGH
EXAM
```

SAMPLE OUTPUT 0

```
Sentence: Spring semester exam is near and exam is tough
Words: 9
Shortest word: is
Keyword count: 2
Vowels: 15
```

Explanation: Only the first letter of the sentence stays uppercase. "exam" appears twice when case is ignored, and "is" is the shortest word.

SAMPLE INPUT 1

```
   The  theory is OK  the theme
the
```

SAMPLE OUTPUT 1

```
Sentence: The theory is ok the theme
Words: 6
Shortest word: is
Keyword count: 2
Vowels: 8
```

Explanation: "is" and "OK" both have 2 letters, so the first one, "is", is printed. "theory" and "theme" are not the same whole word as "the", so the keyword count is 2 (the first and fifth words only).